

Utility Ownership and Residential Electricity Disconnections: An Exploratory Data Analysis (Fall-26 90819 Final Project)

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Introduction:

This Exploratory Data Analysis (EDA) examines 2024 residential electricity disconnections across three utility ownership structures – investor-owned utilities (IOUs), municipal utilities (MUNIs), and electric cooperatives (COOPs) – using data associated with the June 22, 2026, report “New nationwide data reveals utility specific disconnection information”¹ by the Energy & Policy Institute (EPI). The EPI report draws on newly released U.S. Energy Information Administration (EIA) data that provide utility-specific information on residential service disconnections for nonpayment. EPI describes the data as a new national baseline for comparing utility shutoff practices and reports a 93% response rate among electric utilities. Importantly, a reported disconnection is an event rather than necessarily a unique customer: the same customer may be disconnected multiple times during the year. Consequently, a disconnection rate of 50 per 100 customers should not be interpreted as meaning that 50% of customers experienced a disconnection.

As documented in the previously prepared/submitted Dataset Structure and Variables Summary, the underlying workbook² contains eight worksheets comprising utility-level data, subsets, and summary tables. The public-power data includes variables describing state, utility, ownership, customer count, shutoffs, reconnections, final notices, and several corresponding rates and percentiles, while the IOU worksheet contains 169 investor-owned utilities using a somewhat different schema. The present analysis harmonizes the relevant IOU, municipal, and cooperative variables to create a common utility-level dataset containing 1,122 electric utilities: 169 IOUs, 444 MUNIs, and 509 COOPs. The primary outcome is the 2024 disconnection rate, calculated as the number of disconnection events relative to customer count; customer counts and total disconnections are also examined to distinguish differences in utility scale from differences in rates. This distinction is important because, as EPI notes, utilities with large customer bases can record high numbers of total shutoffs without necessarily having the highest disconnection rates.

The objective of the EDA is therefore to determine whether utility ownership type is descriptively associated with 2024 electricity disconnection rates, while also examining

¹ <https://energyandpolicy.org/new-nationwide-data-reveals-utility-specific-disconnection-information/>

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<https://docs.google.com/spreadsheets/d/1WvinnB6o1K7ffaR84NLIUpkuPfPiZrhyd6rgMNSJQoc/edit?usp=sharing>

within-group variability, utility size, geographic variation, and relationships among the continuous variables. The analysis is descriptive rather than causal: observed differences among ownership groups cannot, by themselves, establish that ownership structure causes differences in disconnection practices.

Key Findings:

The EDA identifies a clear descriptive association between utility ownership type and 2024 residential electricity disconnection rates, while also showing substantial variation within each ownership category. Across the 1,122 utilities examined, MUNIs generally exhibit the highest disconnection rates per 100 customers, COOPs occupy an intermediate position, and IOUs have the lowest rates. However, the following seven figures show that this overall pattern is accompanied by right-skewed distributions, substantial overlap among ownership groups, major differences in utility size, geographic variation, and numerous high-rate observations. Consequently, ownership type appears relevant to disconnection outcomes, but it does not by itself explain the considerable utility-to-utility variation observed in the data.

1. Distribution of 2024 Electricity Disconnection Rates

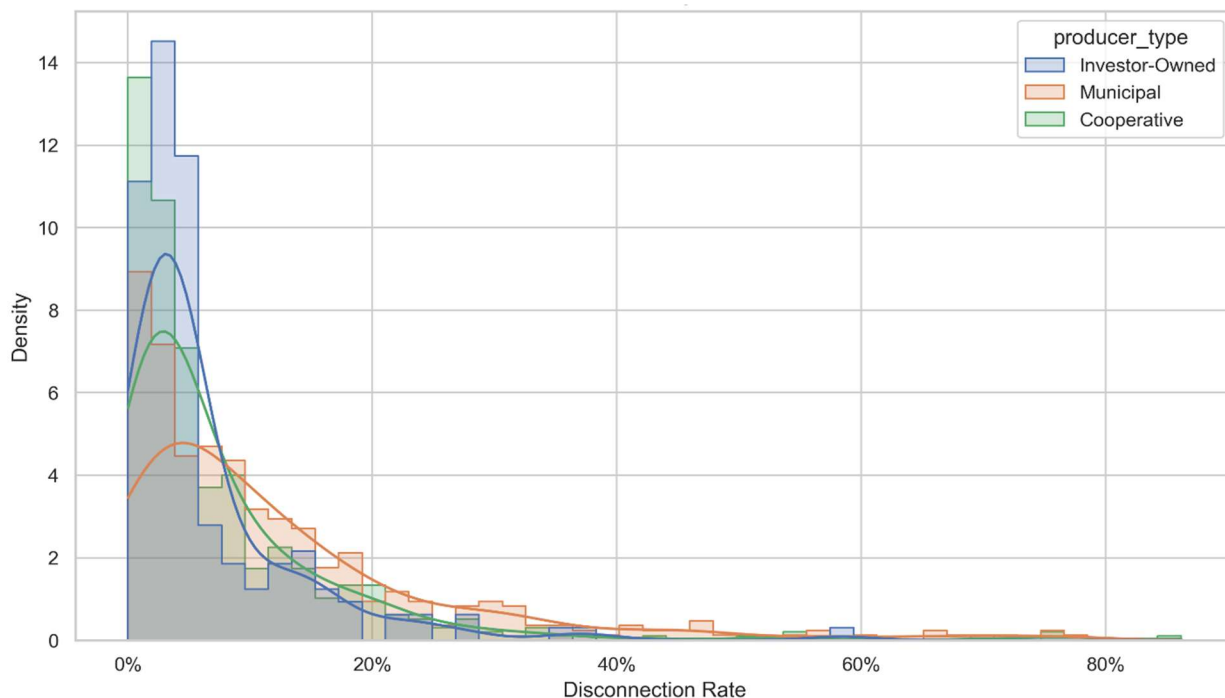


Figure 1 – Distribution of 2024 Electricity Disconnection Rates

The distribution plot provides the broadest view of how disconnection rates vary across the three ownership groups. All three distributions are right-skewed, with most utilities concentrated at relatively low disconnection rates and smaller numbers extending

considerably farther into the upper tail. MUNIs show a distribution shifted toward higher rates relative to IOUs and COOPs, although the three distributions overlap substantially. The long upper tails are also consistent with the 85 potential outliers (21 of which are IOUs, 29 are MUNIs and 35 are COOPs) identified using the $1.5 - IQR$ criterion. These observations were not automatically removed because they may represent genuine utilities with unusually high disconnection rates rather than data errors. The key finding is that ownership groups differ in their overall distributions, particularly for MUNIs, but substantial overlap and long right tails demonstrate considerable variation within every ownership type.

2. Disconnection Rates by Utility Ownership Type

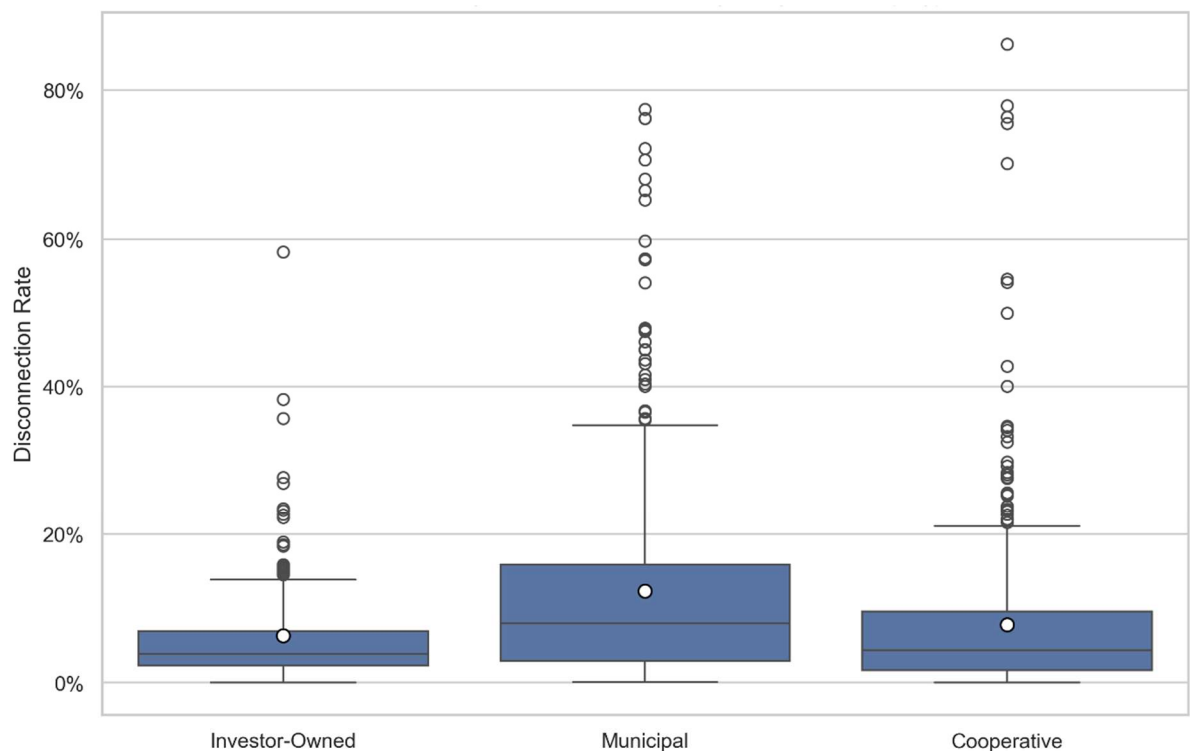


Figure 2 – 2024 Electricity Disconnection Rates by Utility Ownership Type

The box-and-whisker plot makes the differences in central tendency and dispersion more explicit. MUNIs have the highest mean disconnection rate at 12.32 disconnection events per 100 customers³ and highest median at 7.97 disconnection events per 100 customers. COOPs have a mean of 7.81 and median of 4.39, while IOUs have the lowest mean at 6.34 and median at 3.88. The mean exceeds the median by approximately 4.35 disconnection events per 100 customers for MUNIs, 3.42 for

³ the rates are most precisely interpreted as disconnection events per 100 customers, rather than the percentage of unique customers who experienced a disconnection, because an individual customer may potentially experience more than one disconnection event during the year.

COOPs, and 2.47 for IOUs, providing additional evidence of right-skewness. MUNIs also have the greatest dispersion, with an IQR of 13.00 disconnection events per 100 customers, compared with 7.95 for COOPs and 4.64 for IOUs. Nevertheless, the IQRs overlap by approximately 4.01 disconnection events per 100 customers for IOUs and MUNIs, 4.64 for IOUs and COOPs, and 6.70 for MUNIs and COOPs. The key finding is therefore not simply that MUNIs have higher typical rates, but that the differences between ownership groups coexist with substantial within-group variability⁴ and between-group overlap⁵.

3. Utility Size by Ownership Type

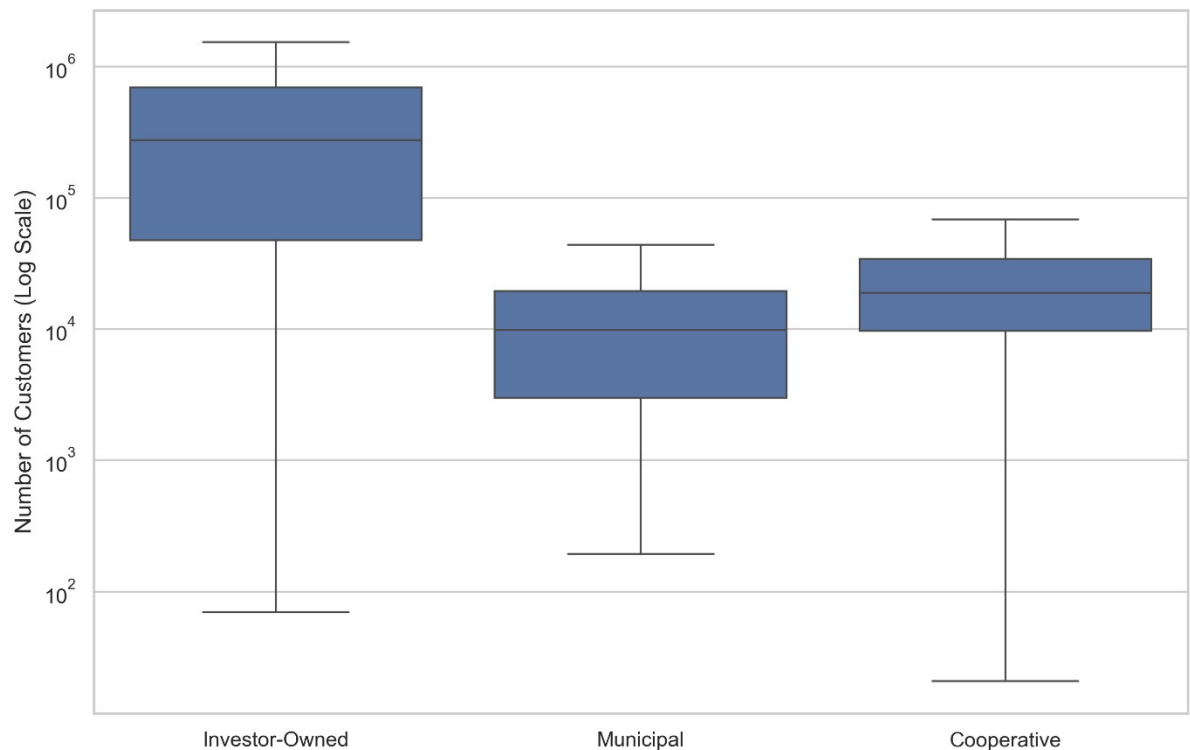


Figure 3 – Utility Size by Ownership Type

Utility size differs dramatically across the three ownership structures. IOUs have a median customer count of approximately 275,025, compared with approximately 18,897 for COOPs and 9,816 for MUNIs. Their mean customer counts similarly differ substantially: approximately 581,566 for IOUs, 31,672 for COOPs, and 27,735 for MUNIs. The logarithmic scale of the plot makes these differences visible despite customer counts spanning several orders of magnitude. This structural difference is particularly important when interpreting raw numbers of disconnections because a

⁴ how much individual utilities differ from other utilities of the same ownership type

⁵ difference across the three ownership types

larger utility has many more customer exposures and therefore greater opportunity to record disconnection events. The key finding is that IOUs are generally much larger than MUNIs and COOPs, making rates rather than raw disconnection counts the more meaningful basis for comparing ownership groups.

4. Utility Size and Disconnection Rate

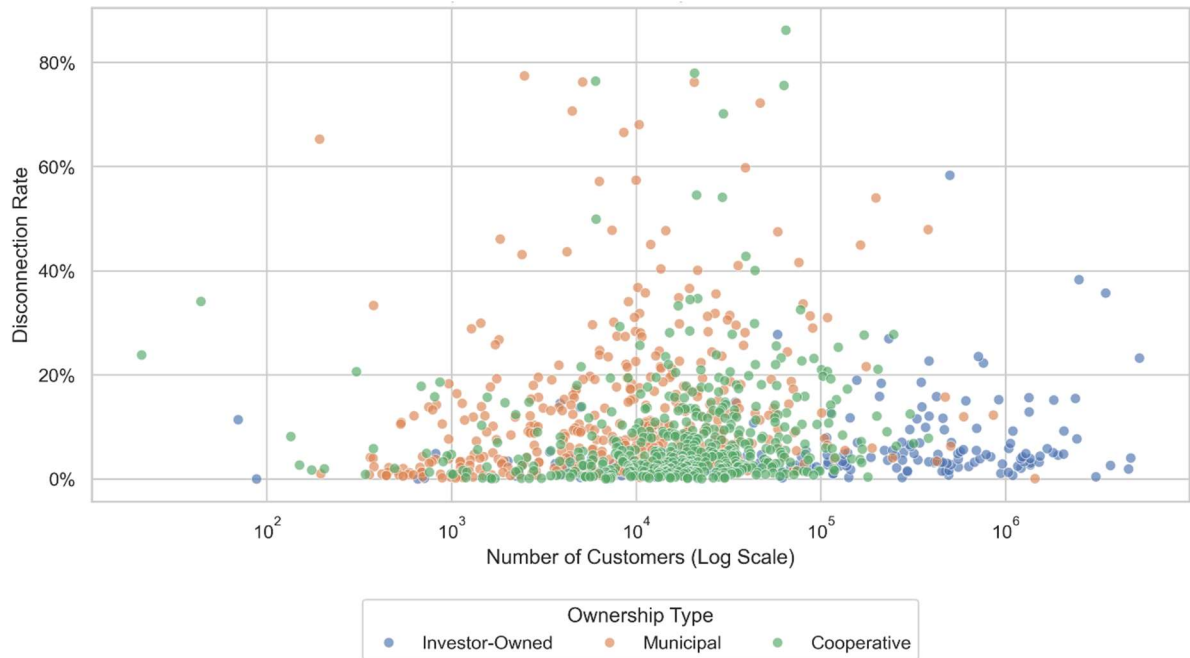


Figure 4 – Utility Size and 2024 Electricity Disconnection Rate

Despite the pronounced differences in utility size, the scatterplot provides little evidence that size itself strongly explains disconnection rates. Across all utilities, the correlation between customer count and disconnection rate is approximately $r = 0.003$, indicating essentially no overall linear relationship. The within-ownership correlations are also relatively weak: approximately 0.225 for IOUs, 0.027 for MUNIs, and 0.128 for COOPs. High and low disconnection rates can therefore be observed among utilities of substantially different sizes. This finding helps separate two characteristics that might otherwise be conflated: ownership groups differ substantially in their typical size, but utility size and disconnection rate are not strongly related. The key finding is that differences in customer base alone do not appear to explain the higher disconnection rates observed among MUNIs or the broader differences among ownership categories.

5. Median Disconnection Rates by State and Ownership Type

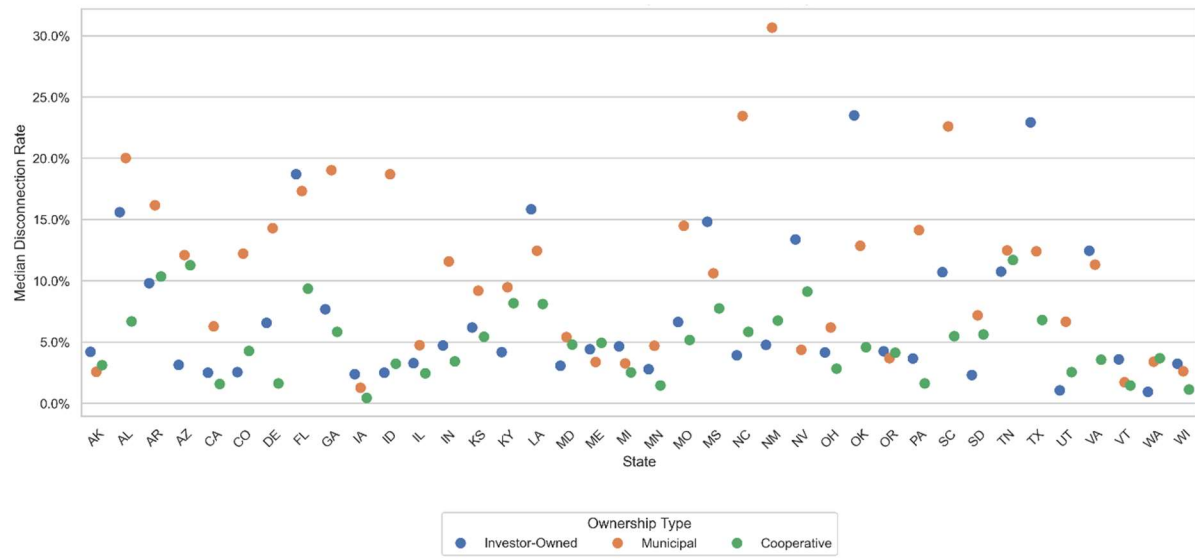


Figure 5 – Median Disconnection Rates by State and Ownership Type

The state-level comparison introduces an important geographic dimension to the ownership results. 38 states contain utilities from all three ownership groups, allowing the median disconnection rates of IOUs, MUNIs, and COOPs to be compared within a shared state context. The figure shows considerable state-to-state variation, as well as differences among ownership types within states, rather than a single uniform national pattern in which one ownership category has the highest median everywhere. This suggests that the national ownership results exist alongside potentially important geographic differences. State regulatory policies, customer characteristics, economic conditions, utility practices, and other state-specific circumstances may contribute to these patterns, although this EDA does not test those explanations. The key finding is that geography matters: the national ownership differences should not be interpreted as uniform across states, and ownership is unlikely to be the only characteristic associated with disconnection rates.

6. Mean Utility Rate versus Aggregate Disconnection Rate

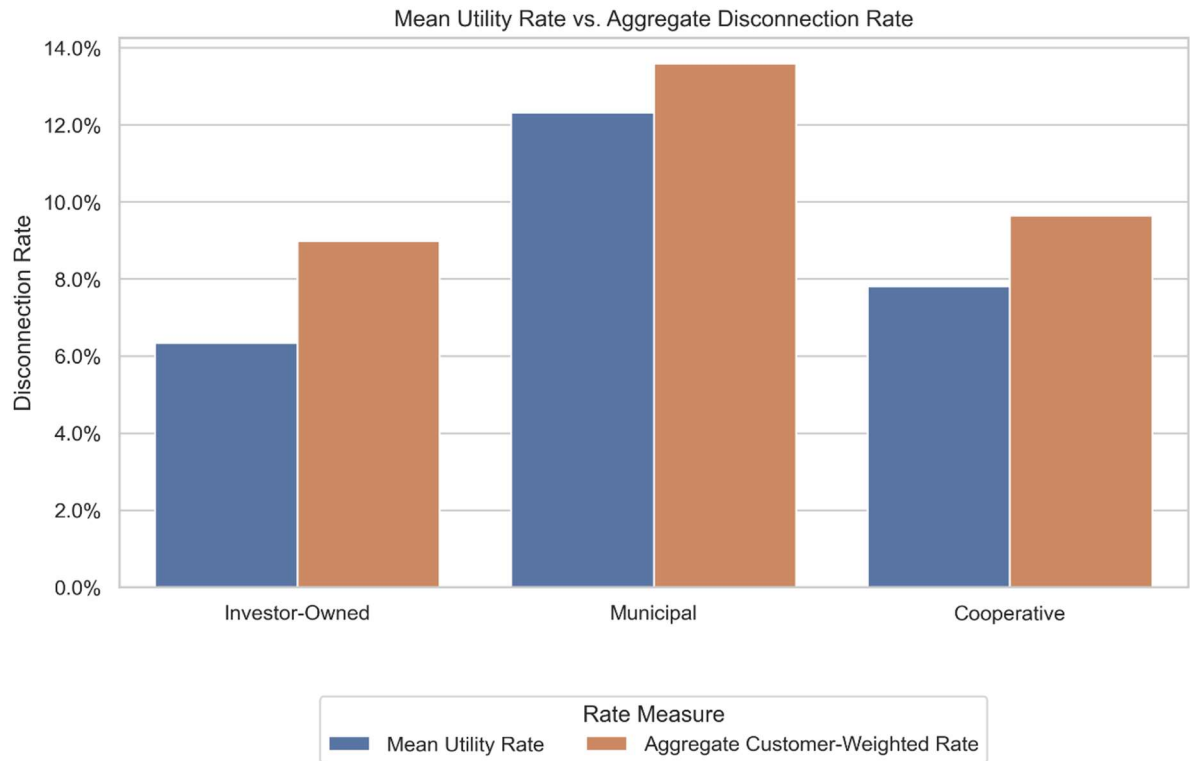


Figure 6 – Mean Utility Rate vs Aggregate Disconnection Rate

Comparing the unweighted mean utility rate with the aggregate customer-weighted rate provides two complementary perspectives on each ownership group. When every utility receives equal weight, the mean rates are 6.34 disconnection events per 100 customers for IOUs, 12.32 for MUNIs, and 7.81 for COOPs. When total disconnections are instead divided by total customers within each ownership group, the aggregate disconnection events per 100 customers rise to 8.98, 13.59, and 9.65, respectively. The fact that the aggregate rate exceeds the unweighted mean for all three groups indicates that larger utilities collectively receive greater influence when customer exposure is incorporated. Importantly, however, the ownership ordering does not change under the alternative calculation: MUNIs remain highest, COOPs second, and IOUs lowest. The key finding is that utility size affects the magnitude of the group-level estimates but does not explain away the principal descriptive ownership pattern.

7. Correlations Among Continuous Variables

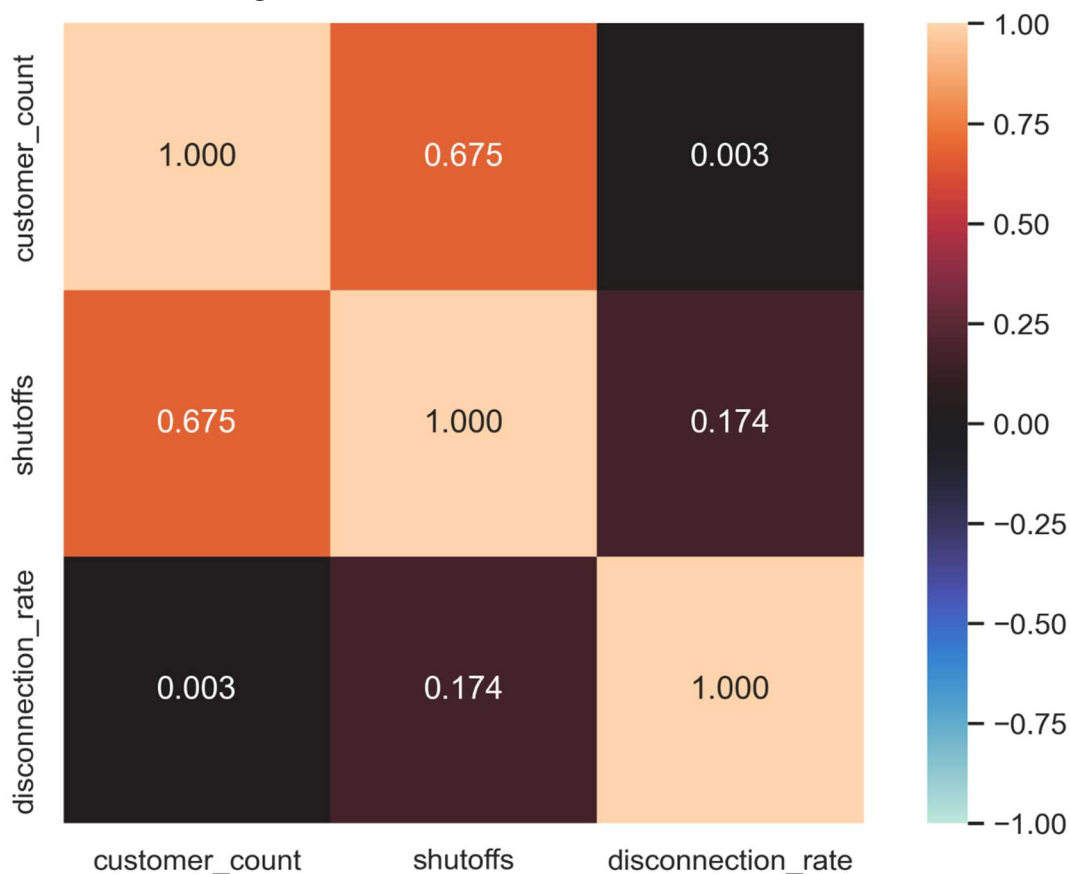


Figure 7 – Correlation Matrix of Continuous Variables

The correlation matrix further clarifies the distinction between the number and rate of disconnections. Customer count and total disconnections have a relatively strong positive correlation of approximately $r = 0.675$, which is expected because utilities serving more customers have greater exposure to potential disconnection events. In contrast, customer count and disconnection rate have essentially no overall linear relationship ($r = 0.003$). Thus, large utilities tend to record more disconnections in absolute terms, but they do not necessarily disconnect at higher rates. The key finding is that raw disconnection totals are strongly influenced by utility scale, whereas normalized disconnection rates largely remove this size relationship, reinforcing the decision to use disconnection rate as the primary outcome for comparing ownership types.

Overall Finding:

Collectively, the seven figures above support the conclusion that utility ownership is descriptively associated with 2024 electricity disconnection rates. MUNIs consistently

have the highest mean and median disconnection events per 100 customers, COOPs generally occupy an intermediate position, and IOUs have the lowest rates. The same ordering persists when customer exposure is incorporated through aggregate rates. At the same time, the results qualify this relationship in several important ways: all three rate distributions are right-skewed; 85 observations are identified as potential outliers; the ownership distributions and IQRs overlap substantially; utility size varies dramatically among ownership types but has almost no overall linear relationship with disconnection rate; and median rates vary across states.

The appropriate interpretation is therefore not that ownership type determines a utility's disconnection rate, but that the 2024 data reveal a meaningful descriptive association between the two. Considerable variation remains within each ownership category, suggesting that other utility-specific and geographic characteristics are also relevant. Because the analysis covers only 2024 and does not control for state regulatory policies, customer socioeconomic characteristics, electricity prices, arrearages, local economic conditions, or individual utility collection practices, it cannot establish causality or determine whether the observed ownership differences would persist after accounting for these factors. This EDA instead establishes the principal patterns in the data and identifies ownership, geography, and utility-specific variation as useful dimensions for further investigation.